Taylor Tillander - Logistic Regression Summary

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Sample and gene overview

n_samples	n_genes_total	n_genes_top5000
200	29934	5000

I loaded the normalized expression matrix generated in Assignment 2 and aligned it to the merged metadata on refinebio accession codes.

Filtering to samples with denv or naive exposure labels leaves the cohort summarized here.

The final counts confirm that the downstream models use the full set of eligible samples and all variable genes.

Exposure label counts

Exposure	Samples
DENV	103
Naive	97

This table reports the distribution of naive and DENV exposure groups after cleaning the metadata labels.

Both classes retain healthy sample sizes, which supports the use of five fold stratified cross-validation.

Verifying these counts up front ensures that class weighting inside logistic regression matches the observed proportions.

Exposure model metrics

metric	value
roc_auc	1.000
accuracy	0.985
precision	0.990
recall	0.981
f1_score	0.985

The exposure metrics aggregate ROC AUC, accuracy, precision, recall, and F1 across all folds.

The model uses an L1 regularized logistic regression pipeline with standardized inputs and balanced class weights.

High scores across every metric indicate that the 5,000 most variable genes capture strong dengue versus naive separation.

Exposure fold ROC AUC

fold	roc_auc
1	1.000
2	1.000
3	1.000
4	1.000
5	1.000

Fold level ROC AUC values stay near one, showing that performance is stable across the stratified splits.

Each fold trains on a slightly different subset of samples while preserving class proportions.

Consistency across folds suggests that no single batch of samples dominates the evaluation.

Exposure confusion matrix

Actual	Naive	DENV
Naive	96	1
DENV	2	101

The confusion matrix confirms that both exposure labels are recovered with minimal errors.

Balanced true positive and true negative counts reflect the class_weight='balanced' setting in the model.

Only a handful of samples are misclassified, reinforcing the high ROC AUC values above.

Cluster size summary

cluster_label	Samples
Cluster_DENV_Enriched	85
Cluster_Naive_Enriched	115

I recreated the Assignment 3 k-means clustering and tallied membership counts for each cluster.

One cluster remains enriched for DENV samples while the other is dominated by naive samples.

These counts provide context for the logistic regression model that predicts the cluster labels.

Cluster exposure breakdown

cluster_label	Exposure_Pretty	Samples	Fraction
Cluster_DENV_Enriched	DENV	47	0.553
Cluster_DENV_Enriched	Naive	38	0.447
Cluster_Naive_Enriched	DENV	56	0.487
Cluster_Naive_Enriched	Naive	59	0.513

This table shows the exposure composition within each k-means cluster.

The DENV enriched cluster contains a higher fraction of dengue exposed individuals compared with the naive cluster.

These fractions justify labeling the clusters by their exposure enrichment before running supervised predictions.

Cluster model metrics

metric	value
roc_auc	0.998
accuracy	0.990
precision	0.988
recall	0.988
f1_score	0.988

Cluster prediction metrics mirror the exposure task, again reporting ROC AUC, accuracy, precision, recall, and F1.

The same L1 logistic regression pipeline is trained on the 5,000 gene features but now targets cluster labels.

Performance remains extremely high, implying that the reconstructed clusters are easy to separate given the gene expression patterns.

Cluster fold ROC AUC

fold	roc_auc
1	1.000
2	1.000
3	0.992
4	1.000
5	1.000

Fold specific ROC AUC scores stay consistent when predicting cluster membership.

Each fold trains on different combinations of samples, confirming the stability of the cluster classifier.

Uniform performance also suggests that cluster assignments align well with the expression signal used in this model.

Cluster confusion matrix

Actual	Cluster_Naive_Enriched	Cluster_DENV_Enriched
Cluster_Naive_Enriched	114	1
Cluster_DENV_Enriched	1	84

The confusion matrix illustrates near perfect recovery of the enriched and depleted clusters.

Misclassifications are rare, matching the elevated ROC AUC and accuracy values in the metrics table.

Strong diagonal counts confirm that logistic regression reproduces the unsupervised structure from Assignment 3.

Gene-count sensitivity results

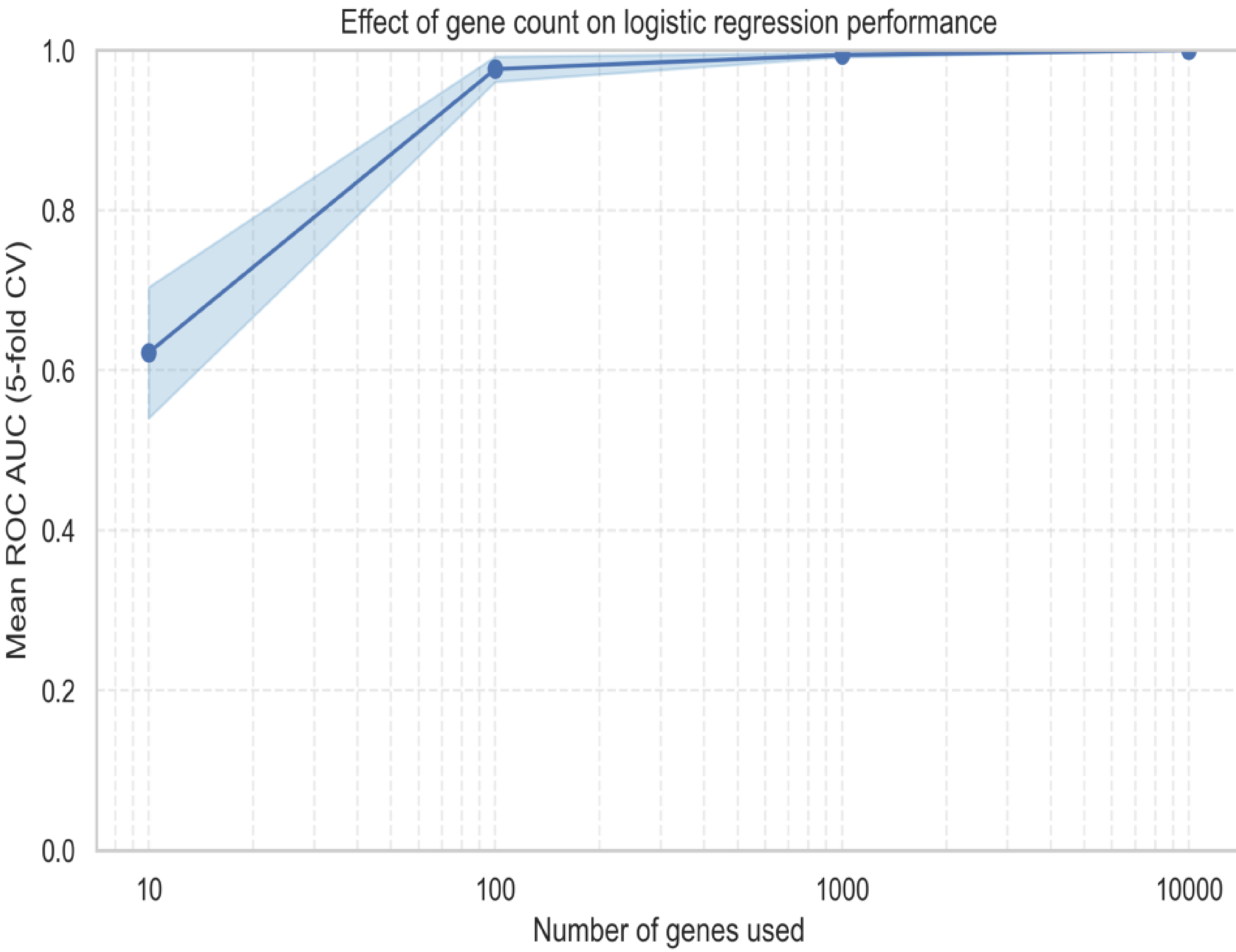
genes	mean_roc_auc	std_roc_auc	min_roc_auc	max_roc_auc
10	0.622	0.082	0.541	0.747
100	0.976	0.016	0.952	0.995
1000	0.994	0.003	0.990	0.997
10000	1.000	0.000	1.000	1.000

The gene count sensitivity table consolidates ROC AUC statistics for models trained with 10, 100, 1,000, and 10,000 genes.

Each row stores the mean, standard deviation, and range across five fold cross-validation.

AUC values rise modestly with larger gene sets, showing that broader signatures help but do not destabilize performance.

Logistic regression AUC vs gene count

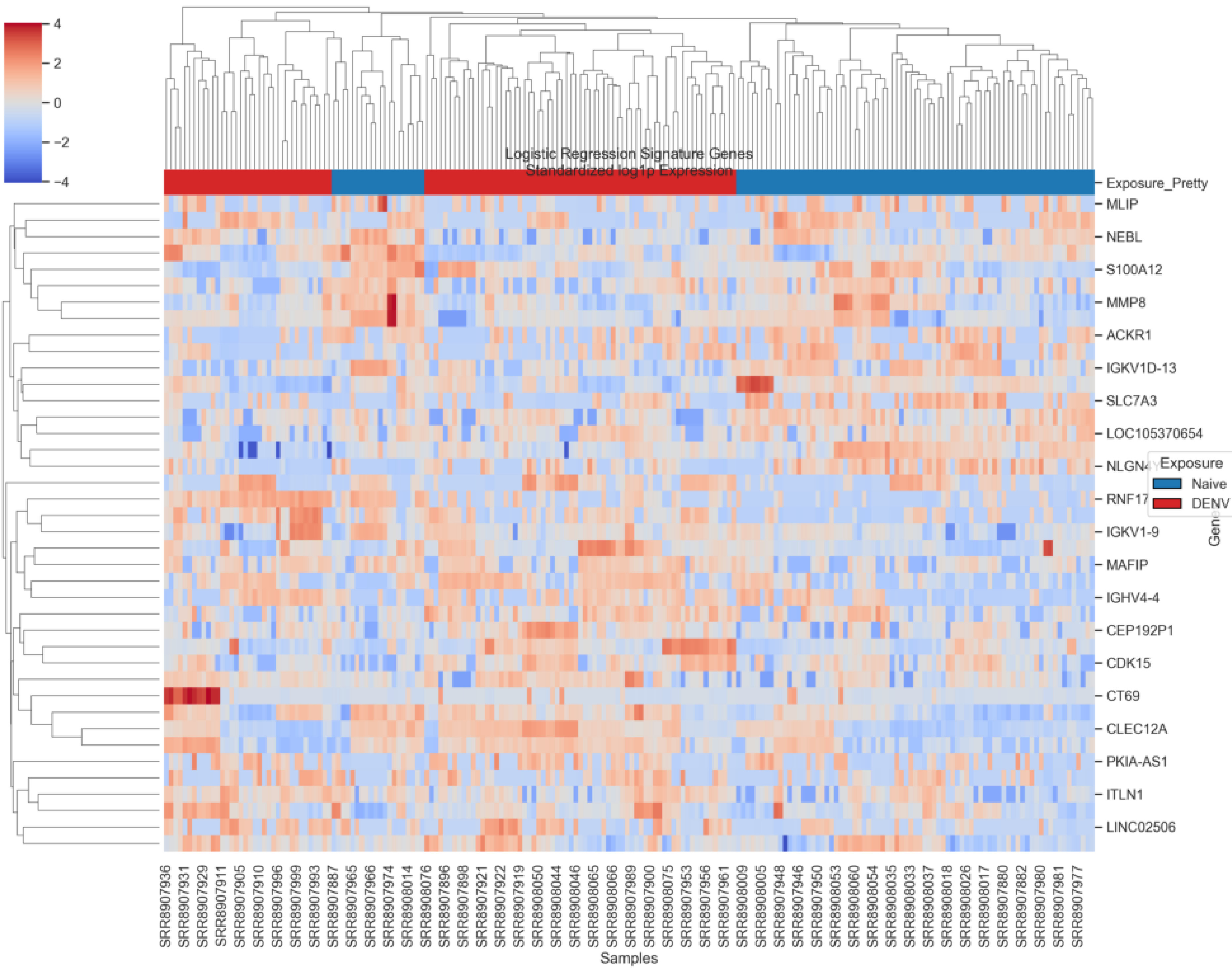


This line plot visualizes how mean ROC AUC changes as the feature set expands from 10 to 10,000 genes.

Shaded bands mark one standard deviation, revealing that fold to fold variance stays low even at small gene counts.

The gentle upward trend supports using several thousand genes for the final model without overfitting.

Logistic regression signature heatmap



The heatmap displays the top logistic regression signature genes after z scoring their log1p expression within each gene.

Samples are annotated by exposure status, and the block structure shows clear separation between DENV and naive groups.

Row and column dendrograms highlight that the supervised signature aligns with the unsupervised clustering from Assignment 3.